Literature Status: Milman's Multi-Integral Norm Question

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Status. **literature_already_answered**. Bizeul–Klartag's 2025 Corollary 1.7 gives the missing M-estimate and explicitly identifies it as completing the conjecture attributed to V. Milman in Skarmogiannis's paper.

Original source and question. Skarmogiannis, *On a multi-integral norm defined by weighted sums of log-concave random vectors*, arXiv:2208.06365 [1], asks in the Introduction (PDF pages 2–3) whether, for a centrally symmetric convex body $K \subset \mathbb{R}^n$, the norm

$$\|\mathbf{t}\|_{K^s,K} = \int_K \cdots \int_K \left\| \sum_{j=1}^{s} t_j x_j \right\|_K dx_1 \cdots dx_s$$

is equivalent to $\|\mathbf{t}\|_2$ up to a factor polylogarithmic in n. For a volume-one isotropic image K_*, Theorem 1.1 and the universal lower bound reduce the question to

$$\sqrt{n}\, M(K_*) L_{K_*} \le C(\log n)^b. \tag{1}$$

The source says immediately after this reduction on PDF page 3, and again in Remark 4.2 on PDF page 9, that (1) remained open.

Supporting answer. Bizeul and Klartag, *Distances between non-symmetric convex bodies: optimal bounds up to polylog*, arXiv:2510.20511 [2], prove in Corollary 1.7 (PDF page 6) that every convex body A in covariance-one isotropic position satisfies

$$M(A) \le C \frac{\psi_n \sqrt{\log n}}{\sqrt{n}}. \tag{2}$$

Their equation (10) records the known estimate $\psi_n \le C\sqrt{\log n}$, so (2) yields

$$M(A) \le C \frac{\log n}{\sqrt{n}}. \tag{3}$$

Normalization and identification. Skarmogiannis uses the volume-one convention $\mathrm{Cov}(K_*) = L_{K_*}^2 I$. Set $A = K_*/L_{K_*}$. Then A is covariance-one isotropic and, by homogeneity of the Minkowski functional,

$$M(A) = L_{K_*} M(K_*).$$

Therefore (3) is exactly (1), with $b = 1$. Substitution into Skarmogiannis's Theorem 1.1, whose remaining loss is $(\log n)^5$, gives the affirmative answer

$$c\|\mathbf{t}\|_2 \leq \|\mathbf{t}\|_{K^s,K} \leq C(\log n)^6 \|\mathbf{t}\|_2 \tag{4}$$

for all $s \geq 1$, after the source's volume normalization.

This is a separate-paper explicit answer: on PDF page 7, immediately after Corollary 1.7, Bizeul–Klartag cite Skarmogiannis and state that their corollary completes, up to polylogarithmic factors, the conjecture attributed to V. Milman on multi-integral norms.

Scope and search evidence. Equation (4) settles the qualitative polylogarithmic-equivalence question and provides exponent 6. It does not assert that this exponent is optimal. The run's four cheap indexes had no prior record for arXiv:2208.06365 or arXiv:2510.20511. Exact-title and exact-$M(K)$ searches located the 2025 preprint; its Corollary 1.7 and explicit citation of the source establish that this is known literature rather than a new proof produced by the run.

References

[1] N. Skarmogiannis, *On a multi-integral norm defined by weighted sums of log-concave random vectors*, arXiv:2208.06365 (2022), Mathematika 69 (2023).

[2] P. Bizeul and B. Klartag, *Distances between non-symmetric convex bodies: optimal bounds up to polylog*, arXiv:2510.20511 (2025).